

4 (long) Figure 3(a) shows a field effect transistor (FET) configured as a source-follower amplifier.

(a) Draw the small-signal equivalent circuit for mid-band frequencies. [4]

(b) Derive expressions for the small-signal voltage gain, the input impedance and the output impedance. [8]

(c) Evaluate the quantities derived in (b) given that the small-signal parameters of the FET are $g_m = 10 \text{ mA V}^{-1}$ and $r_d = 20 \text{ k}\Omega$, and R_1 is $10 \text{ M}\Omega$ and R_2 is $5 \text{ k}\Omega$. [4]

(d) Due to electrical interference from the mains power supply, a low frequency noise signal at 50 Hz is induced at the drain node of the FET circuit. The effects of this noise can be modelled by including a small-signal noise source V_N at the drain of the circuit as illustrated in Fig. 3(b), and by setting v_{in} to zero volts.

(i) Draw the small-signal equivalent circuit. [4]

(ii) Derive an expression for the small-signal voltage gain $|v_{out}/V_N|$ and calculate the value of this gain given the parameters listed in (c). [4]

(e) A load resistor R_L of $100 \text{ }\Omega$ is connected across the output terminals of Fig. 3(b). To filter out low frequency noise signals, a capacitor C_{out} is placed at the output of the circuit. Explain whether C_{out} should be placed in series or in parallel with R_L . Using the values obtained in part (c), calculate the value of C_{out} required to achieve a 3 dB cut-off frequency of 200 Hz. [6]

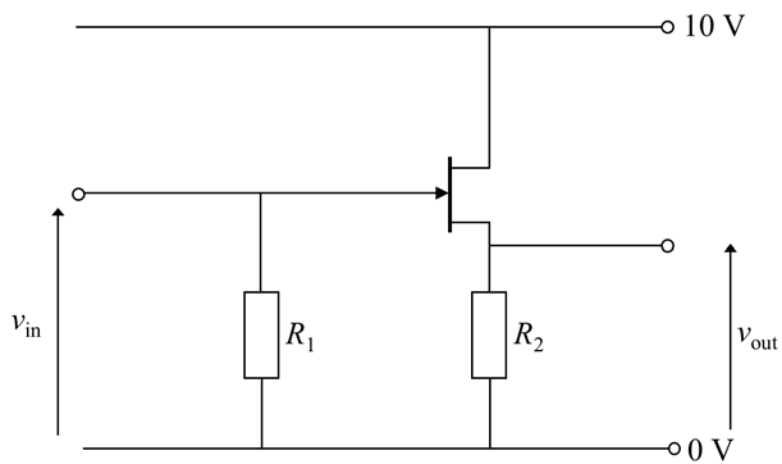


Fig. 3(a)

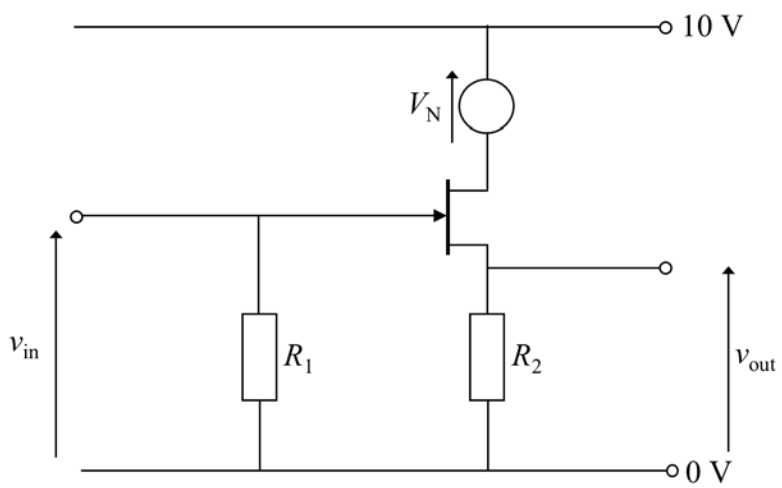


Fig. 3(b)